Configuration Field

Magnetic field due to a single circular loop. Closed-form solution due to Biot-Savart law.

$$B^{\ell}(\mathbf{r}, t) = I(t) \cdot \left[g_{\rho}^{\ell}(\mathbf{r})\ \hat{\rho} + g_{z}^{\ell}(\mathbf{r})\ \hat{z} \right]$$

Where each loop's geometry (radius and height) determine the geometric factors g_r and g_z, and $\mathbf{r} = \rho\hat{\rho} + z\hat{z}$ The total field of a configuration, consisting of several circular loops is given by the superposition of the singular fields

$$B(\mathbf{r}, t) = I(t) \begin{bmatrix} \sum_{\ell} g_{\rho}^{\ell}(\mathbf{r})\ \hat{\rho} \\ \sum_{\ell} g_{z}^{\ell}(\mathbf{r})\ \hat{z} \end{bmatrix} = I(t) \begin{bmatrix} g_{\rho}(\mathbf{r})\ \hat{\rho} \\ g_{z}(\mathbf{r})\ \hat{z} \end{bmatrix} \tag{1}$$

such that having several loops with different parameters boils down to summing their geometric factors for every component.

Eddy Currents

A magnetic flux through an area induces eddy currents. For a changing magnetic field in the z direction, eddy currents are expected to be generated in the ϕ direction (due to symmetry of the problem)

> **Skin Depth** ›
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> Magnetic diffusion time is $\tau_d \approx \mu_0 \sigma d^2 \approx 4\ [\mu s]$ for $d = 42\ [\text{mm}]$ and $\sigma \approx 1.9 \times 10^7\ [\text{S/m}]$. The rise time of the pulse is $\tau_p \approx 10\ [\mu s] \gg \tau_d$ which means the magnetic field has time to fully penetrate the disc such that we are in the quasi-static limit.

$$J_{\phi}(\mathbf{r}, t) = \sigma E_{\phi}(\mathbf{r}, t) = -\frac{\sigma}{2\pi\rho} \frac{\partial \Phi_z(\mathbf{r}, t)}{\partial t} \tag{2}$$

This is a direct result form Faraday's law

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

The change in magnetic flux, calculated explicitly using equation (1), gives

$$\begin{aligned}
\frac{\partial \Phi_z(\mathbf{r},t)}{\partial t} &= \dot{I}(t) \iint\limits_A \mathbf{B}(\mathbf{r},t)\cdot \hat{z}\, dA \\
&= \dot{I}(t) \left[\int\limits_0^{rho} g_z(\rho',z)\, \rho' d\rho' \right] \\
&\equiv 2\pi\, \dot{I}(t) G_z(\mathbf{r})
\end{aligned}$$

Plugged back into equation (2) we have a formula for the eddy currents generated

$$J_\phi(\rho,t) = -\frac{\sigma}{\rho} G_z(\mathbf{r})\, \dot{I}(t) \tag{3}$$

Lorentz Force

Now that we have a formula for the Eddy currents we can calculate the force generated between them and the radial magnetic field, namely

$$\begin{aligned}
\mathbf{f}(\mathbf{r},t) &= \mathbf{J} \times \mathbf{B} \\
&= J_\phi \Big[B_\rho (\hat{\phi} \times \hat{\rho}) + B_z (\hat{\phi} \times \hat{z}) \Big] \\
&= J_\phi \Big[B_\rho (-\hat{z}) + B_z (\hat{\rho}) \Big]
\end{aligned}$$

We are interested in the force distribution in the z direction since it generates the pressure pulse

$$f_z(\mathbf{r},t) = -J_\phi(\mathbf{r},t) \cdot B_\rho(\mathbf{r},t) \tag{4}$$

Plugging into (4) equations (3) and (1) respectively yields

$$\begin{aligned}
f_z(\mathbf{r},t) &= \frac{\sigma}{\rho} G_z(\mathbf{r}) g_\rho(\mathbf{r}) \dot{I}(t) I(t) \\
&= K(\mathbf{r})\, \frac{d}{dt}\Big(I^2(t) \Big)
\end{aligned} \tag{5}$$

$$K(\mathbf{r}) = \frac{\sigma}{2\rho} g_\rho(\mathbf{r}) \int\limits_0^{\rho} g_z(\mathbf{r}) \rho' d\rho' \tag{5*}$$

Assuming force gradients along the z-direction are negligible, our quantity of interest is the radial force distribution (surface force distribution), namely

$$f_z^{\text{surf}}(\rho, t) = \int_{h_1}^{h_2} f(\mathbf{r}, t) dz \tag{6}$$

where $h_2 - h_1$ is the height of the disc. The total force on the disc is then

$$F_z(t) = 2\pi \int_0^R f_z^{\text{surf}}(\rho, t)\ \rho d\rho \tag{7}$$

with R being the disc's radius.

- Goal 1 - produce a surface plot (circular) of $f_z^{\text{surf}}(\rho, t)$ with an h_1 slider, setting $h_2 = h_1 + d$ and an output of the $F_z(t)$ calculation.
- Goal 2 - from $f_z^{\text{surf}}(\rho, t)$, or $F_z(t)$ (whichever makes better sense) find the generated $p(z, t)$ which travels in the upward z-direction throughout the membrane.